TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	74.9 ± 2.26
Correlation Stability	92.3 ± 0.64
Distribution Consistency	81.7 ± 1.19
Entropy Preservation	85.7 ± 1.68
Mean Z Score Drift	0.2 ± 0.03
Categorical Fidelity	93.5 ± 3.61
Privacy Risk	47.8 ± 1.17
Data Density	100.0 ± 0.02
Coverage Score	25.4 ± 7.17
Feature Accuracy	89.7 ± 3.23
Novelty Score	0.5 ± 0.01

TabuLLM: AI Benchmark Report

Dataset Audit: indian_liver_patient_dataset.csv

1. Dataset Portrait

Attribute	Value
Rows	583
Columns	11
Numeric features	10
Categorical features	1
Missing cells (%)	0.1%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	20%
Generation % (*15)	100%
Join context (*16)	True

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
Age	44.746	49.550	+4.804	10.7%	Warn
TB	3.299	2.605	-0.694	21.0%	Warn
DB	1.486	1.110	-0.376	25.3%	Drift
Alkphos	290.576	263.509	-27.068	9.3%	OK
Sgpt	80.714	54.246	-26.468	32.8%	Drift
Sgot	109.911	59.462	-50.449	45.9%	Drift
TP	6.483	7.022	+0.538	8.3%	OK
ALB	3.142	3.573	+0.431	13.7%	Warn
A/G Ratio	0.947	1.012	+0.065	6.8%	OK
Selector	1.286	1.000	-0.286	22.3%	Warn

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	74.9 ± 2.26
Correlation Stability (*2)	92.3 ± 0.64
Distribution Consistency (*3)	81.7 ± 1.19
Entropy Preservation (*4)	85.7 ± 1.68
Mean Z Score Drift (*5)	0.2 ± 0.03
Categorical Fidelity (*6)	93.5 ± 3.61
Privacy Risk (*7)	47.8 ± 1.17
Data Density (*8)	100.0 ± 0.02
Coverage Score (*9)	25.4 ± 7.17

Feature Accuracy (*10)	89.7 ± 3.23
Novelty Score (*11)	0.5 ± 0.01

[1] (*12) Algorithm: Contextual constraint or model parameter.

[2] (*13) Folds: Contextual constraint or model parameter.

[3] (*14) Context %: Contextual constraint or model parameter.

[4] (*15) Generation %: Contextual constraint or model parameter.

[5] (*16) Join context: Contextual constraint or model parameter.

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6615384615384616	0.6809116809116809	0.6495726495726497	-0.011965811965811968	
F1	0.7644157232977931	0.7908569192643798	0.7540828769816448	-0.010332846316148203	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6794871794871795	0.7008547008547007	0.6871794871794872	0.007692307692307709	
F1	0.7873449447804652	0.8138080337856742	0.7912437115471289	0.0038987667666663653	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5589743589743594	0.7094017094017094	0.5623931623931624	0.003418803418803488	
F1	0.5587432119380917	0.8299804995124879	0.5666709977747274	0.007927785836635814	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7068376068376068	0.7065527065527065	0.7068376068376068	0	
F1	0.8279091624529537	0.8254490704372873	0.8279091624529537	0	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7025641025641025	0.7037037037037037	0.7119658119658119	0.009401709401709401	
F1	0.7997903571549948	0.8110388310022216	0.8055437439276218	0.005753386772627089	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6863247863247863	0.6780626780626781	0.6871794871794872	0.00085470085470086	
F1	0.7971299319768493	0.8004843598101447	0.7994688418501645	0.00233890987331513	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7059829059829059	0.7094017094017094	0.7094017094017093	0.00341880341880340 67	
F1	0.8128002697439015	0.8208247442126564	0.8170699344614771	0.00426966471757563 9	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6871794871794872	0.7150997150997151	0.6752136752136753	-0.011965811965812001	
F1	0.7950296303884821	0.8338805970149252	0.7824643549627998	-0.012565275425683198	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6743589743589744	0.6638176638176638	0.688034188034188	0.01367521367521366	
F1	0.7821343746683586	0.7624939573720062	0.7931006744059634	0.010966299737604856	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6965811965811967	0.7037037037037037	0.7102564102564103	0.013675213675213682	
F1	0.8047862802917571	0.8173160210508152	0.8113772861498324	0.006591005858075372	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	65	Female	0.7	0.1	187	16	18	6.8	3.3	0.9
2	62	Male	10.9	5.5	699	64	100	7.5	3.2	0.74
3	62	Male	7.3	4.1	490	60	68	7.0	3.3	0.89
4	58	Male	1.0	0.4	182	14	20	6.8	3.4	1.0
5	72	Male	3.9	2.0	195	27	59	7.3	2.4	0.4
6	46	Male	1.8	0.7	208	19	14	7.6	4.4	1.3
7	26	Female	0.9	0.2	154	16	12	7.0	3.5	1.0
8	29	Female	0.9	0.3	202	14	11	6.7	3.6	1.1
9	17	Male	0.9	0.3	202	22	19	7.4	4.1	1.2
10	55	Male	0.7	0.2	290	53	58	6.8	3.4	1.0
11	57	Male	0.6	0.1	210	51	59	5.9	2.7	0.8
12	72	Male	2.7	1.3	260	31	56	7.4	3.0	0.6
13	64	Male	0.9	0.3	310	61	58	7.0	3.4	0.9
14	74	Female	1.1	0.4	214	22	30	8.1	4.1	1.0
15	61	Male	0.7	0.2	145	53	41	5.8	2.7	0.87
16	25	Male	0.6	0.1	183	91	53	5.5	2.3	0.7
17	38	Male	1.8	0.8	342	168	441	7.6	4.4	1.3
18	33	Male	1.6	0.5	165	15	23	7.3	3.5	0.92
19	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8
20	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8

TabuLLM: AI Benchmark Report

B.1 Audit Trace: Fold 1 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	46	Male	10.2	4.2	232	58	140	7.0	2.7	0.6
2	50	Male	2.6	1.2	415	407	576	6.4	3.2	1.0
3	52	Male	2.7	1.4	251	20	40	6.0	1.7	0.39
4	62	Male	1.2	0.4	195	38	54	6.3	3.8	1.5
5	46	Female	4.7	2.2	310	62	90	6.4	2.5	0.6
6	42	Male	11.1	6.1	214	60	186	6.9	2.8	2.8
7	54	Male	3.5	1.2	200	45	60	7.1	3.5	0.9
8	36	Male	2.2	1.0	220	50	70	6.8	3.4	0.9
9	48	Female	1.5	0.6	190	35	45	7.0	3.1	0.8
10	34	Female	2.1	0.9	210	40	50	6.7	3.3	1.0
11	45	Female	4.0	1.5	300	65	80	7.5	3.6	1.1
12	52	Male	1.8	0.7	230	55	65	6.9	3.2	0.8
13	33	Male	7.1	3.7	196	622	497	6.9	3.6	1.09
14	38	Male	2.2	1.0	310	119	42	7.9	4.1	1.0
15	64	Male	3.0	1.4	248	46	40	6.5	3.2	0.9
16	40	Female	0.9	0.2	285	32	27	7.7	3.5	0.8

B.2 Audit Trace: Fold 2 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	33	Male	2.1	1.3	480	38	22	6.5	3.0	0.8
2	45	Male	2.4	1.1	168	33	50	5.1	2.6	1.0
3	29	Male	0.8	0.2	156	12	15	6.8	3.7	1.1
4	65	Male	0.9	0.2	170	33	66	7.0	3.0	0.75
5	43	Male	0.8	0.2	192	29	20	6.0	2.9	0.9
6	42	Male	8.9	4.5	272	31	61	5.8	2.0	0.5
7	52	Female	2.5	1.0	200	50	45	7.0	3.8	1.2
8	38	Female	3.0	1.5	220	60	50	7.5	4.0	1.3
9	39	Female	3.5	1.8	230	65	55	7.8	4.2	1.4
10	55	Female	2.8	1.2	240	70	65	8.0	4.5	1.5
11	61	Female	3.2	1.6	220	55	50	7.4	4.1	1.2
12	53	Female	2.3	1.1	210	60	55	7.1	3.7	1.3
13	50	Female	1.0	0.3	191	22	31	7.8	4.0	1.0
14	52	Female	0.6	0.1	194	10	12	6.9	3.3	0.9
15	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
16	58	Male	0.8	0.2	298	33	59	6.2	3.1	1.0

B.3 Audit Trace: Fold 3 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	64	Male	1.1	0.4	201	18	19	6.9	4.1	1.4
2	48	Male	4.5	2.3	282	13	74	7.0	2.4	0.52
3	21	Male	3.9	1.8	150	36	27	6.8	3.9	1.34
4	62	Male	10.9	5.5	699	64	100	7.5	3.2	0.74
5	50	Male	0.9	0.3	194	190	73	7.5	3.9	1.0
6	40	Male	0.7	0.2	176	28	43	5.3	2.4	0.8
7	52	Female	2.5	1.0	250	50	60	6.8	4.0	1.1
8	41	Male	3.0	1.5	300	70	80	7.5	3.5	0.9
9	48	Male	2.0	0.8	220	45	50	7.0	3.6	1.1
10	70	Male	4.5	2.0	310	55	65	8.0	4.2	1.0
11	39	Male	2.8	1.2	280	60	70	7.2	3.9	1.0
12	26	Female	0.7	0.3	160	15	20	5.5	2.6	0.8
13	44	Female	1.9	0.6	298	378	602	6.6	3.3	1.0
14	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
15	53	Male	0.9	0.4	238	17	14	6.6	2.9	0.8
16	28	Male	0.8	0.3	190	20	14	4.1	2.4	1.4

B.4 Audit Trace: Fold 4 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	27	Male	1.0	0.2	205	137	145	6.0	3.0	1.0
2	14	Male	1.4	0.5	269	58	45	6.7	3.9	1.4
3	42	Female	0.8	0.2	182	22	20	7.2	3.9	1.1
4	75	Male	14.8	9.0	1020	71	42	5.3	2.2	0.7
5	48	Male	5.0	2.6	555	284	190	6.5	3.3	1.0
6	38	Female	2.6	1.2	410	59	57	5.6	3.0	0.8
7	46	Female	2.5	1.0	300	50	60	7.0	3.5	1.2
8	39	Male	1.8	0.7	250	60	65	6.8	3.4	1.0
9	41	Male	2.3	0.9	220	55	45	6.7	3.3	1.0
10	37	Male	2.0	1.0	280	65	75	7.2	3.7	1.1
11	52	Male	2.8	1.2	300	70	80	7.0	3.5	1.0
12	54	Male	3.5	1.8	350	80	85	7.4	3.9	1.2
13	42	Male	0.7	0.2	197	64	33	5.8	2.4	0.7
14	69	Male	0.9	0.2	215	32	24	6.9	3.0	0.7
15	53	Male	1.6	0.9	178	44	59	6.5	3.9	1.5
16	49	Male	1.3	0.4	206	30	25	6.0	3.1	1.06

B.5 Audit Trace: Fold 5 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	42	Male	11.1	6.1	214	60	186	6.9	2.8	2.8
2	38	Female	0.6	0.1	165	22	34	5.9	2.9	0.9
3	70	Male	1.3	0.4	358	19	14	6.1	2.8	0.8
4	30	Male	1.6	0.4	332	84	139	5.6	2.7	0.9
5	26	Male	6.8	3.2	140	37	19	3.6	0.9	0.3
6	48	Female	0.8	0.2	218	32	28	5.2	2.5	0.9
7	60	Male	3.5	1.0	400	70	80	7.5	3.8	1.3
8	48	Male	2.5	1.2	310	55	65	7.2	3.6	1.2
9	54	Male	1.9	0.5	290	52	62	7.1	3.5	1.1
10	62	Male	3.0	1.1	350	65	75	7.4	3.7	1.3
11	46	Male	2.3	0.9	280	49	53	6.7	3.4	1.1
12	55	Female	1.7	0.6	230	39	44	6.5	3.1	1.1
13	33	Male	2.1	0.7	205	50	38	6.8	3.0	0.7
14	22	Female	6.7	3.2	850	154	248	6.2	2.8	0.8
15	48	Male	5.8	2.5	802	133	88	6.0	2.8	0.8
16	84	Female	0.7	0.2	188	13	21	6.0	3.2	1.1

B.6 Audit Trace: Fold 6 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	54	Male	0.8	0.2	218	20	19	6.3	2.5	0.6
2	34	Male	6.2	3.0	240	1680	850	7.2	4.0	1.2
3	51	Male	2.9	1.3	482	22	34	7.0	2.4	0.5
4	26	Male	1.9	0.8	180	22	19	8.2	4.1	1.0
5	38	Female	2.6	1.2	410	59	57	5.6	3.0	0.8
6	55	Male	18.4	8.8	206	64	178	6.2	1.8	0.4
7	50	Male	1.8	0.7	220	45	55	6.2	3.2	1.0
8	45	Female	2.1	0.9	210	38	44	6.8	3.3	1.1
9	62	Male	0.6	0.1	230	55	65	6.4	3.4	1.2
10	60	Male	3.5	1.8	250	55	70	7.3	3.9	1.2
11	37	Female	3.5	1.5	300	35	50	6.5	2.8	0.8
12	61	Female	2.1	0.9	180	22	30	5.9	2.4	0.7
13	50	Male	0.9	0.3	194	190	73	7.5	3.9	1.0
14	49	Male	0.6	0.1	218	50	53	5.0	2.4	0.9
15	18	Male	0.8	0.2	228	55	54	6.9	4.0	1.3
16	30	Female	0.7	0.2	194	32	36	7.5	3.6	0.92

B.7 Audit Trace: Fold 7 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	65	Male	0.8	0.1	146	17	29	5.9	3.2	1.18
2	50	Male	1.7	0.8	331	36	53	7.3	3.4	0.9
3	60	Male	2.0	1.1	664	52	104	6.0	2.1	0.53
4	52	Female	0.6	0.1	194	10	12	6.9	3.3	0.9
5	35	Female	0.6	0.2	180	12	15	5.2	2.7	nan
6	72	Male	1.7	0.8	200	28	37	6.2	3.0	0.93
7	55	Male	3.5	1.0	300	50	60	7.0	3.5	1.0
8	63	Male	2.1	0.9	400	45	55	6.8	3.4	1.1
9	72	Male	4.0	1.5	500	60	70	7.5	3.8	1.2
10	48	Male	3.0	1.2	350	65	50	6.9	3.3	0.9
11	66	Male	5.2	2.0	600	75	80	7.8	4.0	1.3
12	58	Male	3.8	1.4	400	70	65	7.2	3.7	1.1
13	13	Female	0.7	0.2	350	17	24	7.4	4.0	1.1
14	54	Female	23.2	12.6	574	43	47	7.2	3.5	0.9
15	63	Male	0.9	0.2	194	52	45	6.0	3.9	1.85
16	46	Male	0.6	0.2	290	26	21	6.0	3.0	1.0

B.8 Audit Trace: Fold 8 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	72	Male	0.7	0.1	196	20	35	5.8	2.0	0.5
2	48	Female	0.8	0.2	142	26	25	6.0	2.6	0.7
3	20	Female	0.6	0.2	202	12	13	6.1	3.0	0.9
4	70	Female	0.9	0.3	220	53	95	6.1	2.8	0.68
5	22	Female	6.7	3.2	850	154	248	6.2	2.8	0.8
6	60	Male	8.9	4.0	950	33	32	6.8	3.1	0.8
7	45	Male	2.0	0.8	250	55	60	7.5	3.9	1.2
8	41	Male	3.5	1.5	300	70	75	8.0	4.0	1.3
9	48	Male	2.5	1.2	260	72	78	8.1	4.3	1.3
10	44	Male	2.1	0.7	250	53	58	7.7	4.0	1.2
11	37	Male	2.5	1.1	300	60	70	7.0	3.5	1.0
12	54	Male	3.2	1.5	400	80	90	7.5	3.9	1.1
13	48	Male	0.7	0.2	208	15	30	4.6	2.1	0.8
14	60	Male	22.8	12.6	962	53	41	6.9	3.3	0.9
15	27	Male	1.0	0.3	180	56	111	6.8	3.9	1.85
16	28	Female	0.8	0.2	309	55	23	6.8	4.1	1.51

B.9 Audit Trace: Fold 9 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	60	Male	2.1	1.0	191	114	247	4.0	1.6	0.6
2	41	Male	2.7	1.3	580	142	68	8.0	4.0	1.0
3	66	Male	0.6	0.2	100	17	148	5.0	3.3	1.9
4	38	Female	0.7	0.1	152	90	21	7.1	4.2	1.4
5	37	Male	0.8	0.2	195	60	40	8.2	5.0	1.5
6	32	Male	15.9	7.0	280	1350	1600	5.6	2.8	1.0
7	54	Male	3.5	1.5	250	60	70	7.0	3.5	1.0
8	58	Male	4.0	1.8	300	70	80	6.8	3.4	1.3
9	37	Male	2.5	1.1	320	55	75	7.1	3.6	1.0
10	60	Male	3.2	1.4	290	65	85	6.5	3.3	1.1
11	48	Male	2.8	1.3	310	50	60	7.0	3.4	1.2
12	39	Female	1.4	0.6	230	38	48	6.4	3.1	1.1
13	26	Male	2.0	0.9	195	24	65	7.8	4.3	1.2
14	47	Male	2.7	1.3	275	123	73	6.2	3.3	1.1
15	74	Male	1.0	0.3	175	30	32	6.4	3.4	1.1
16	48	Male	3.2	1.6	257	33	116	5.7	2.2	0.62

B.10 Audit Trace: Fold 10 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	33	Male	2.0	1.0	258	194	152	5.4	3.0	1.25
2	47	Male	0.9	0.2	192	38	24	7.3	4.3	1.4
3	55	Male	14.1	7.6	750	35	63	5.0	1.6	0.47
4	54	Male	2.2	1.2	195	55	95	6.0	3.7	1.6
5	60	Female	1.4	0.7	159	10	12	4.9	2.5	1.0
6	65	Male	0.9	0.2	170	33	66	7.0	3.0	0.75
7	52	Male	3.1	1.4	400	80	50	8.0	3.8	0.95
8	48	Male	2.0	0.9	270	65	55	7.8	4.1	1.15
9	30	Male	1.8	0.7	240	50	38	7.1	3.4	1.0
10	60	Male	3.5	1.6	320	75	50	7.4	4.2	1.1
11	50	Male	1.9	0.8	300	55	40	7.0	3.5	1.0
12	62	Male	2.4	1.1	340	80	60	8.1	4.3	1.2
13	54	Female	5.5	3.2	350	67	42	7.0	3.2	0.8
14	43	Male	22.5	11.8	143	22	143	6.6	2.1	0.46
15	57	Male	4.5	2.3	315	120	105	7.0	4.0	1.3
16	65	Male	0.7	0.2	199	19	22	6.3	3.6	1.3